Seeding QDArchive

(5 / 10 / 15 ECTS Seminar and / or Project)

Dirk Riehle, FAU Erlangen
Professorship for Open-Source Software

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QDArchive

QDArchive

A web service for researchers to publish and archive qualitative data

- With an emphasis on qualitative data analysis (QDA) files

Qualitative data = interview transcripts, research articles, audio/video, etc.

- QDA files = Structured data that capture the meaning in the qualitative data

Both qualitative data files and QDA files are useful for RAG creation

- And LLM training

Seeding QDArchive

QDArchive is new and under active development (still a prototype)

Chicken and egg problem = how to attract researchers?

Answer = seed QDArchive with available open qualitative data

This project is to lay the groundwork for seeding QDArchive

Project Course

Seminar / Project Timeline

Part 1: **Data acquisition** (5 ECTS)

- ~~Until March 15th~~ Until April 17th

Part 2: **Data classification** (5 ECTS)

- ~~Until April 12th~~ Until May 29th

Part 3: **Data analysis** (5 ECTS)

- ~~Until May 31st~~ Until July 17th

Ways of Participating and Acquiring ECTS

- 5 ECTS Seminar (generic name “Applied Software Engineering Seminar”)
  - Acquisition only
  - Classification only
  - Analysis only
- 10 ECTS Project (generic name “Applied Software Engineering Project”)
  - Acquisition + classification
  - Classification + analysis
  - ~~Acquisition + analysis~~
- 15 ECTS
  - Seminar + Project

Coordination Meetings

Acquisition start = Feb 13 (Fri), 4pm CET

- Project meetings on Feb 20 (Fri), Feb 26 (Thu), Mar 6 (Fri), Mar 13 (Fri)
- Always at 4pm CET in our public lecture hall

Classification start = Mon Mar 16, 4pm CET

- Three weekly meetings Mondays Mar 23, 30, Apr 6 at 4pm

Analysis start = Fri Apr 17, 5pm CET

- Relaxed, until end of teaching time, schedule TBD then

Part 1: Acquisition

Part 1 Step 1

Find as much qualitative **research** projects on the web as you can

- Start with finding qualitative data research projects
  - Can be identified by containing Analysis Data files (QDA files)
  - Or may simply have a description that says qualitative (data) research
- Try to find **Analysis Data files** (not just qualitative data)
  - Everything can be interpreted as qualitative data
  - We don't want you (to try) downloading arxiv
- Collect as much metadata as possible with the qualitative data
  - **The data must have an open license**
  - So ensure you get the license
- **Basic heuristic: Just download the research project folder (all files)**

A Qualitative (Data) Research Project is a Web Folder

QDA Research Project Data Set (website or folder or archive)

Analysis Data file (a.k.a. QDA file)
- The key data structure (file) we are interested in; result of a researcher analyzing a domain
- Possible extensions: qdpx, mx24, etc. see file extensions spreadsheet
- Number of files: 0 or 1 (most likely), but also possible that there is more than 1 file
- Analysis files may contain none, some, or all primary data files

Primary Data files
- The input data to a researcher's analysis, for example, interview transcripts or research papers
- Possible extensions: pdf, doc, docx, txt, jpg, etc. (anything goes)
- Number of files: 0..n (if 0, hopefully the primary data files are included in the analysis data file)
- Primary data files may be included in an analysis data file and may also be standalone files in the research project

Additional Data files
- Other additional files the researcher considered important (that are not analysis or primary data files)
- Possible extensions: pdf, doc, docx, txt, jpg, etc. (anything goes)
- Number of files: 0..n (no expectations)
- Please also collect them

DR

Part 1 Analysis Data File (QDA File) Types

An analysis data file is structured data that

- A scientists creates when interpreting primary data like interview transcripts

We have a particular interest in qualitative data analysis (QDA) files, e.g.

- qdpx (called the REFI format, see https://www.qdasoftware.org/

Spreadsheet with known file types / extensions

- Please add to it if you find new information

Part 1 Primary Data File Types

A primary data file is a file that contains any form of qualitative data like

- Interview transcripts or research articles

File types include

- txt, PDF, RTF, DOCX, …

Part 1 Additional Data Files

In the scope of a qualitative research project there can be other files that are

- Not analysis or primary data files
- But something the researcher thinks belongs to the project

Part 1 Qualitative Data Repositories

Qualitative research projects are often archived in repositories

- Data sources to start with

Part 1 Open Data Licenses

- Any Creative Commons license
- Many others…

It will be more difficult to find a license

- No license = close proprietary do not use

Part 1 Step 2

Build a pipeline to download all this data

- Both files as well as metadata
  - Put metadata into an associated database (exportable to a CSV file)
- The database schema will be evolved during the project
  - Source, source link, license information, project scope, …
- If you can’t fill a particular field, leave it empty

Part 1 Step 3

Operate the pipeline to create your own archive

Part 1 Ongoing

Report about technical challenges with data (not with programming)

Part 1 Submission / Delivery

Work is due by March 15th

Tag your git version as part-1-release

Fill in a form, which asks about

- Download link for database
- Link to folder with downloaded files

Part 2: Classification

Part 2 Step 1

Merge all student databases into one (for yourself) to work on

- Identify and remove duplicates, if possible

Part 2 Step 2

Develop a classifier for the available data

- Uses both the base data (the file) and the metadata

Use the ISIC Rev. 5 standard for a hierarchical classification taxonomy

- Go down two levels, i.e. divisions (not just sections)

Also consider creating tags for searching

Part 2 Step 3

Run the classifier on your data

Part 2 Step 4

Report about the resulting statistics

- How much qualitative data was found
- The distribution of sections and divisions in the data

Part 2 Ongoing

Report about technical challenges with data (not with programming)

Part 3: Analysis

Part 3

Still thinking about it; in contrast to Part 1 and Part 2 these may be individual projects

One-off project: Feeding the archives into QDArchive

Thank you! Any questions?

dirk.riehle@fau.de – https://oss.cs.fau.de

dirk@riehle.org – https://dirkriehle.com – @dirkriehle

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